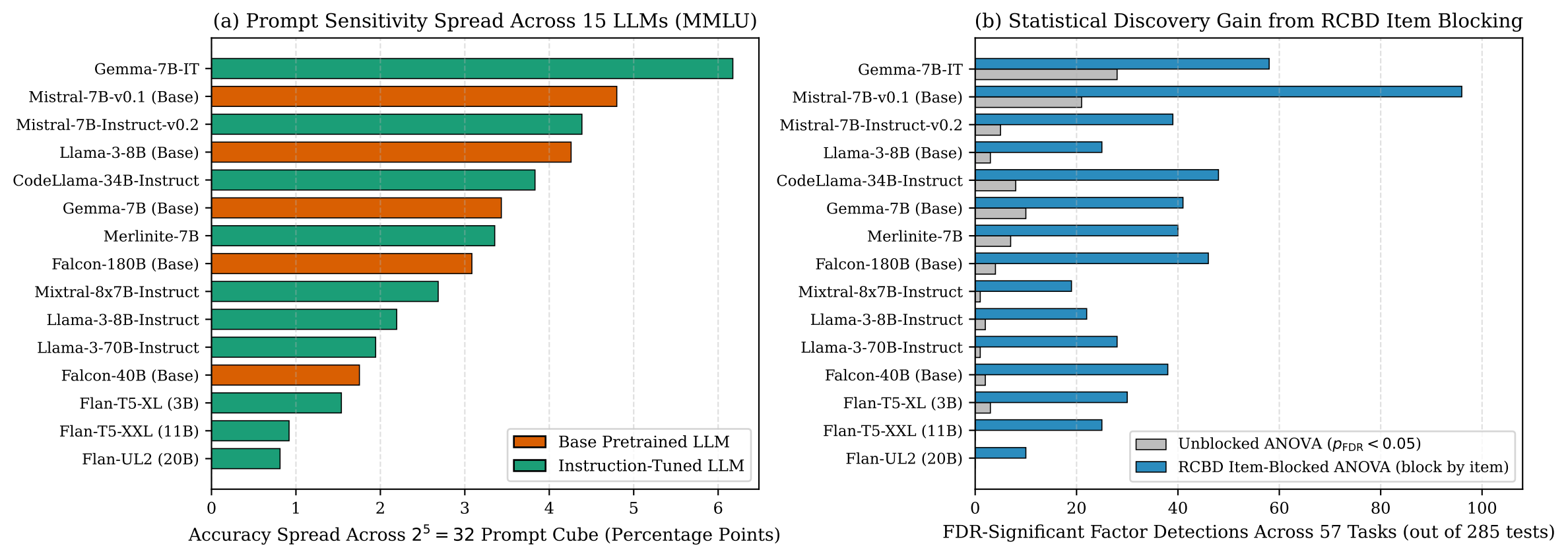
(a) Prompt Sensitivity Spread Across 15 LLMs (MMLU)
Gemma-7B-IT
Mistral-7B-v0.1 (Base)
Mistral-7B-Instruct-v0.2
Llama-3-8B (Base)
CodeLlama-34B-Instruct
Gemma-7B (Base)
Merlinite-7B
Falcon-180B (Base)
Mixtral-8x7B-Instruct
Llama-3-8B-Instruct
Llama-3-70B-Instruct
Falcon-40B (Base)
Flan-T5-XL (3B)
Flan-T5-XXL (11B)
Flan-UL2 (20B)
Base Pretrained LLM
Instruction-Tuned LLM
0
1
2
3
4
5
6
Accuracy Spread Across $2^5 = 32$ Prompt Cube (Percentage Points)
(b) Statistical Discovery Gain from RCBD Item Blocking
Gemma-7B-IT
Mistral-7B-v0.1 (Base)
Mistral-7B-Instruct-v0.2
Llama-3-8B (Base)
CodeLlama-34B-Instruct
Gemma-7B (Base)
Merlinite-7B
Falcon-180B (Base)
Mixtral-8x7B-Instruct
Llama-3-8B-Instruct
Llama-3-70B-Instruct
Falcon-40B (Base)
Flan-T5-XL (3B)
Flan-T5-XXL (11B)
Flan-UL2 (20B)
Unblocked ANOVA ($p_{FDR} < 0.05$)
RCBD Item-Blocked ANOVA (block by item)
0
20
40
60
80
100
FDR-Significant Factor Detections Across 57 Tasks (out of 285 tests)